

TO-BE Cross-Role Process Analysis — Online Food Delivery System

Status: APPROVED Version: 1.0 Project: Online Food Delivery System Artefact Type: TO-BE Cross-Role Business Process Analysis Source Baseline: S1 — Validated Portfolio BRD Depends On: 00_README_BA_Evidence_Pack.md — APPROVED v1.0; AS_IS_Process.md — APPROVED v1.0 Last Reviewed: 15 Aug 2026

1. Purpose

This artefact documents the intended future-state cross-role process for the Online Food Delivery System. Its objectives are to define the end-to-end target order-fulfilment workflow, identify role handoffs, and identify key system-mediated process points. It establishes the target process baseline for later business-rule, state-model, use-case, and acceptance-criteria artefacts, while demonstrating how the future-state process addresses the main limitations identified in the approved AS-IS baseline.

2. TO-BE Process Scope

This target process focuses on the core order and delivery lifecycle involving the Customer, Restaurant, Shipper, and the Online Food Delivery System. Administrator activities, such as partner account approval, are upstream prerequisites and are not part of the active, core order-fulfilment flow.

The core scope covers:

- Customer ordering (browsing, cart)
- Checkout (validation, fee calculation, payment method selection)
- Order confirmation / pending confirmation
- Restaurant preparation
- Ready for Pickup
- Shipper availability / acceptance
- Pickup
- Delivery
- Customer tracking
- Completion
- Review
- Shipper cancellation exception

3. TO-BE Participants & Responsibilities

Customer

Responsible for browsing restaurants and items, adding items to the cart, proceeding through checkout, confirming the order, tracking delivery progress, and submitting post-completion reviews.

Restaurant

Responsible for viewing incoming orders, processing/preparing the food, and updating the order readiness status once preparation is complete.

Shipper

Responsible for viewing available unassigned delivery orders, accepting an eligible delivery, travelling to collect the order, executing the physical delivery, and completing the delivery within the system.

Online Food Delivery System

Responsible for mediating the entire workflow. Explicit responsibilities include presenting restaurant/menu information, capturing the checkout intent, validating the delivery address and determining route/distance via Map APIs, then calculating delivery fees according to defined business rules, creating the order and maintaining its status, exposing eligible orders to Shippers, supporting status/location tracking, and recording lifecycle updates.

4. TO-BE Process Narrative

The target end-to-end process is coordinated through the Online Food Delivery System at its key business handoffs and lifecycle checkpoints. It begins with the Customer browsing menus and submitting an order via checkout, where the System automatically validates the address, calculates delivery fees, and captures the payment choice (COD or QR Payment Simulation). Upon confirmation, the System records the order and makes the new order available to the Restaurant. The Restaurant processes the order and, once prepared, triggers a status update indicating the food is ready for pickup.

The System then exposes this ready, unassigned order to eligible Shippers. A Shipper accepts the order, claims the assignment, and travels to the Restaurant. Upon collection, the Shipper updates the System to indicate the delivery is actively in progress. Throughout this phase, the Customer uses the System to track the order status and Shipper routing. Finally, the Shipper completes the delivery, updating the System to the terminal success state, which then unlocks the Customer's ability to submit ratings and reviews for both the Restaurant and Shipper.

5. TO-BE Process Steps

Step	Primary Participant	Activity	System / Cross-role Handoff	Result / Process State	Evidence Classification	Source
1	Customer	Browses restaurants, selects menu items, and adds to cart.	Interacts with System	Cart populated	BASELINE	S1

Step	Primary Participant	Activity	System / Cross-role Handoff	Result / Process State	Evidence Classification	Source
2	Customer	Proceeds to checkout.	Submits intent to System	Checkout initiated	BASELINE	S1
3	System	Validates delivery address, determines coordinates/routes, calculates delivery cost.	System automation	Validation complete	BASELINE	S1
4	Customer	Selects COD or QR Payment Simulation and confirms the order.	Customer → System	Order submitted	BASELINE	S1
5	System	Creates the order and makes it available to the Restaurant for handling.	System → Restaurant	"Chờ xác nhận"	BASELINE	S1
6	Restaurant	Views the new order and prepares the food.	Reads order from System	Preparation active	BASELINE	S1
7	Restaurant	Selects "Làm xong" when preparation is completed.	Restaurant → System	"Đang lấy món" (Ready for Pickup)	BASELINE	S1
8	System	Exposes the unassigned order to eligible Shippers.	System → Shipper	Order available for assignment	BASELINE	S1
9	Shipper	Views available unassigned orders and accepts a delivery.	Shipper → System	Order assigned to Shipper	BASELINE	S1
10	Shipper	Travels to restaurant, collects the order, and updates status.	Shipper → System	"Đang giao" (Delivering)	BASELINE	S1
11	Customer	Tracks order status and Shipper location/route through web interface.	System → Customer	Visibility provided	BASELINE	S1
12	Shipper	Completes the physical delivery and updates the order status.	Shipper → System	"Hoàn thành" (Completed)	BASELINE	S1
13	Customer	Submits ratings/reviews for the Restaurant and Shipper.	Customer → System	Reviews recorded	BASELINE	S1

6. Process-State Checkpoints

- **Chờ xác nhận (Pending Confirmation)**
 - **Trigger:** Customer confirms/submits the checkout.
 - **State-setting responsibility:** System creates the order with status "Chờ xác nhận".
 - **Next Actor:** Restaurant can view and process the order.
- **Đang lấy món (Ready for Pickup)**
 - **Trigger:** Restaurant marks the food preparation as complete ("Làm xong").
 - **Ownership:** Initiated by Restaurant.
 - **Next Actor:** System exposes the order; Shipper is now able to accept it.
- **Đang giao (Delivering)**
 - **Trigger:** Shipper collects the order and indicates departure.
 - **Ownership:** Initiated by Shipper.
 - **Next Actor:** Shipper controls the delivery progression; Customer gains active transit visibility.

- **Hoàn thành (Completed)**
 - **Trigger:** Shipper successfully finishes the delivery.
 - **Ownership:** Initiated by Shipper.
 - **Next Actor:** Customer is now able to submit reviews.

7. Cross-Role Handoffs

- **Customer → System (Confirmed Checkout)**
 - **Information:** Order submission, address, selected payment method.
 - **Result:** System generates the formal order record.
 - **Classification:** BASELINE
- **System → Restaurant (Order Availability)**
 - **Information:** New order details.
 - **Result:** Restaurant begins preparation.
 - **Classification:** BASELINE
- **Restaurant → System (Readiness Update)**
 - **Information:** Food preparation completed.
 - **Result:** Order state updates to "Đang lấy món", making it eligible for Shipper acceptance.
 - **Classification:** BASELINE
- **System → Shipper (Delivery Opportunity)**
 - **Information:** Unassigned, ready-for-pickup delivery order.
 - **Result:** Shipper evaluates and accepts the delivery.
 - **Classification:** BASELINE
- **Shipper → System (Delivery Acceptance & Updates)**
 - **Information:** Assignment acceptance, delivery-status update after collection, and completion update.
 - **Result:** System records assignment and lifecycle-status updates; delivery tracking information is made available to the Customer as defined by the target requirements.
 - **Classification:** BASELINE
- **System → Customer (Tracking & Status)**
 - **Information:** Order progress and Shipper location.
 - **Result:** Customer gains visibility into active order status and delivery routing information.
 - **Classification:** BASELINE

8. Exception Flow — Shipper Cancels Accepted Delivery

If a Shipper cancels an assigned delivery, the Shipper assignment is cleared and the order returns to the available delivery pool as defined by S1. The exact post-cancellation order-status transition required to make the order eligible for re-acceptance is not explicitly defined in the current baseline and requires clarification against BR-SHIP-02.

9. Business-Rule Touchpoints

The target process intersects with underlying business rules at the following key points:

- **Delivery Feasibility:** The System must validate the delivery distance and calculate fees before allowing checkout confirmation (touches BR-DEL-01, BR-FEE-01, BR-FEE-02).
- **Shipper Eligibility / Assignment:** A Shipper may only accept an order when it is unassigned and marked as ready for pickup (touches BR-SHIP-02). A Shipper cannot hold multiple active delivery orders simultaneously (touches BR-SHIP-01).
- **Post-Completion Review:** The Customer is only permitted to review an order after it has reached the completed state (touches BR-ORDER-01).

10. AS-IS to TO-BE Process Improvements

AS-IS Limitation	TO-BE Process Change	Expected Process Improvement	Classification
Manual Order Recording	Structured digital order capture via checkout	Replaces assumed manual capture with structured platform-based order capture	BASELINE change / DERIVED improvement
Manual Shipper Coordination	Platform-mediated visibility and acceptance	Replaces direct manual Restaurant-to-Shipper coordination with platform-mediated assignment flow	BASELINE change / DERIVED improvement
Limited Delivery Visibility	Order-status and routing tracking	Provides structured delivery-progress visibility through the platform	BASELINE change / DERIVED improvement
Fragmented Operational Data	Platform-based lifecycle and history recording	Centralizes key lifecycle information and reduces information fragmentation	BASELINE change / DERIVED improvement

11. System Responsibility Boundary

System-supported:

- Displaying restaurant and menu data.
- Validating delivery address and determining route/distance via Map APIs; calculating delivery fees and order totals according to defined business rules.
- Generating the order record and persisting status changes.
- Exposing eligible unassigned orders to Shippers.
- Facilitating Customer tracking of order status and Shipper location.
- Recording lifecycle events and capturing post-delivery reviews.

Human/Business actions:

- **Customer:** Browses, adds items, selects payment, confirms checkout, and writes reviews.
- **Restaurant:** Prepares the physical order and triggers the readiness update in the portal.
- **Shipper:** Evaluates/accepts assignments, travels to the restaurant, collects the physical order, executes the delivery, and triggers the completion update.

12. Source Mapping

Process Element	Source	Evidence Classification	Notes
Checkout validation, fee calculation, payment selection	S1	BASELINE	Defined in validated BRD
"Chờ xác nhận" state on creation	S1	BASELINE	Target lifecycle checkpoint
"Đang lấy món" triggered by Restaurant	S1	BASELINE	Target lifecycle checkpoint
Shipper acceptance and "Đang giao" update	S1	BASELINE	Target lifecycle checkpoint
Customer tracking via web interface	S1	BASELINE	Target customer capability
"Hoàn thành" and post-delivery review	S1	BASELINE	Target lifecycle checkpoint
Shipper cancellation return-to-availability flow	S1	BASELINE	Exception defined by validated process

Note: S2 (Original Major Project Report) was reviewed for compatible supporting context but was not materially relied upon for the TO-BE baseline, as S1 provides the definitive target requirement.

13. Open Questions & Not Evidenced

- The following target-process details cannot currently be established from the approved evidence:
- **Post-cancellation status transition:** whether an order cancelled after reaching "Đang giao" must transition back to "Đang lấy món" before becoming eligible for another Shipper — **NOT EVIDENCED / REQUIRES CLARIFICATION.**
 - **Restaurant confirmation semantics:** whether "Chờ xác nhận" requires a distinct Restaurant confirmation action and whether a separate intermediate status exists before "Đang lấy món" — **NOT EVIDENCED / REQUIRES CLARIFICATION.**
 - precise timing SLA or timeout for Restaurant confirmation — NOT EVIDENCED
 - automated reassignment timing or Shipper response timeout — NOT EVIDENCED
 - notification channel details (e.g., push notifications vs. manual refresh) — NOT EVIDENCED (S1 lists push notifications as Future Scope)
 - exact tracking coordinate refresh frequency — NOT EVIDENCED
 - cancellation penalty policy for Shippers — NOT EVIDENCED

14. Analysis Summary

The TO-BE target process replaces fragmented, manual coordination with a structured, platform-mediated order lifecycle connecting the Customer, Restaurant, and Shipper. Through structured digital order capture, rule-constrained lifecycle progression, platform-mediated delivery acceptance, and integrated tracking, the target process is intended to replace the manual cross-role coordination identified in the AS-IS baseline.

This approved TO-BE process serves as an upstream baseline for `Business_Rules_Catalogue.md` and subsequent system-behaviour and requirements artefacts.